

A Novel ϕ -Based Numeral System Derived from 6-Basis Vectors

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Abstract

We introduce a numeral system based on a 6-basis vector representation, in which each component represents an exponent of the golden ratio (ϕ). Unlike traditional positional numeral systems, increments are defined by integer powers of ϕ , which satisfies $\phi^n = \phi^{n-1} + \phi^{n-2}$, with $\phi^2 = \phi + 1$. Each valid 6-vector corresponds uniquely to an intersection point in Cartesian space. This system has potential applications in computational mathematics, geometric modeling, and theoretical exploration.

1 Introduction

Traditional numeral systems rely on integer bases such as 2, 10, or 16. In contrast, we propose a numeral system based on the golden ratio ϕ , an irrational number with unique algebraic properties. Numbers are represented via 6-basis vectors, each component encoding a power of ϕ . A unique Cartesian intersection criterion defines the validity of these vectors.

2 Theoretical Background

2.1 The Golden Ratio and Recurrence

The golden ratio ϕ is defined by:

$$\phi = \frac{1 + \sqrt{5}}{2},$$

satisfying the recurrence:

$$\phi^n = \phi^{n-1} + \phi^{n-2}, \quad \phi^2 = \phi + 1.$$

2.2 6-Basis Vectors and Intersection

We define six basis vectors in $\mathbb{R}^3$:

$$\text{sixBases} = \{[\phi, 0, 1], [\phi, 0, -1], [0, 1, \phi], [0, -1, \phi], [1, \phi, 0], [-1, \phi, 0]\}.$$

A 6-vector $\mathbf{v} = [v_0, \dots, v_5]$ is valid if it yields a unique intersection point via:

$$\text{sixToCartesian}(\mathbf{v}) = \sum_{i=0}^5 v_i \cdot \text{sixBases}[i].$$

We specifically consider partial vectors of the form $[a, ?, c, ?, e, ?]$ where unknown components satisfy:

$$\begin{aligned} b &= \frac{2e}{\phi} - a, \\ d &= -c, \\ f &= -e. \end{aligned}$$

Thus, the intersection coordinates become:

$$X = 4e, \quad Y = 2c, \quad Z = 2 \left(a - \frac{e}{\phi} \right).$$

3 Methodology

Our procedure includes two key steps:

1. Completing partial vectors using:

$$b = \frac{2e}{\phi} - a, \quad d = -c, \quad f = -e.$$

2. Mapping vectors to Cartesian points via:

$$\text{sixToCartesian}(\mathbf{v}) = \sum_{i=0}^5 v_i \cdot \text{sixBases}[i].$$

The consistency is validated against expected coordinates.

4 Results and Discussion

Preliminary computational tests demonstrate the robustness of this system. Visualization tools confirm the unique encoding of intersection points. The numeral representation introduces a non-integer radix based explicitly on ϕ , offering a fresh perspective on numeral system structures.

5 Conclusion

We have developed a novel numeral system based on a 6-basis vector framework using ϕ . The strict Cartesian intersection criterion guarantees unique number representation. Future research includes refining the arithmetic framework and exploring applications in computational mathematics, cryptography, and quantum computing.

Acknowledgments

[Include acknowledgments here.]

Appendix: On Arithmetic in the ϕ -Based System

The arithmetic (addition and subtraction) rules for this numeral system remain an open challenge. Unlike standard numeral systems, arithmetic in this ϕ -based system involves bidirectional carry propagation, as demonstrated by the example:

$$2 \neq \phi^2 + \phi^{-1} \approx 3.236,$$

but rather:

$$2 = \phi + \phi^{-2} \approx 2.0.$$

Future work must rigorously define arithmetic operations, normalization, and stability criteria to ensure the consistency and uniqueness of numeral representations.

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